

Contact

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(LinkedIn)

Top Skills

SOLIDWORKS

Ansys Mechanical

Manufacturing Operations
Management

Languages

Marathi (Native or Bilingual)

Kannada (Native or Bilingual)

Spanish (Limited Working)

English (Full Professional)

Hindi (Native or Bilingual)

Certifications

Exploring Aerospace Coatings and Composites: Material Choices, Design Considerations, and Performance

Introduction to Generative AI

Quadcopter Workshop

Manufacturing Process Simulation using Ansys LS-DYNA and Ansys Forming

Boeing National Aeromodelling Festival

Honors-Awards

ATL STUDENT INNOVATOR
AWARD

Best Delegate Award [Virtual MUN by AWMUN]

All India Rank 50 in GATE Exam (Aerospace Engineering)

Best Project Presentation Award – Innovation Gallery 2026

Samarth .R. Kulkarni

Aero/Astro & Biz. Adm. | AIR-50 GATE'26 AE | former Intern @ SDSC-SHAR(ISRO), Sriharikota | former Intern @ R&DE(DRDO), Pune

Mumbai, Maharashtra, India

Summary

Hi, my name is Samarth! I'm a final-year Aerospace Engineering student at Amity University Maharashtra, Mumbai. I am passionate about merging diverse scientific fields to uncover the extraordinary and exploring space Structures & Materials and Space Sustainability research. Driven by a research interest in the development of innovative aerospace systems that balance sustainability with operational efficiency. Firm believer in making a tangible impact on daily lives through my work.

Current research interests include reconfigurable space structures, advanced composite materials, In-Space Manufacturing, and Structural optimization for advanced aerospace systems. Proficient in structural analysis (FEA), CAD, and CFD.

Following the path inspired by the visionary legacies of Dr. A.P.J. Abdul Kalam and Sir Ratan Tata, I aim to contribute to technologies that are not only advanced but also meaningful for society.

Experience

ISRO - Indian Space Research Organization

Summer Research Intern

June 2025 - July 2025 (2 months)

Sriharikota

internship at the Indian Space Research Organisation (ISRO), Satish Dhawan Space Centre SHAR, with a focus on structural analysis about S200 Solid Stage Boosters (LVM3). The role involves exposure to materials and structural testing, and simulation methodologies relevant to launch vehicle systems.

Team AeroSustain

Team Lead

January 2023 - June 2025 (2 years 6 months)

Mumbai, Maharashtra, India

Active Member of Team AeroSustain, a dynamic club focused on innovation in drones, RC planes, model rocketry, and robotics. Collaborating on diverse projects and competing on the national and international stage, pushing the boundaries of technology and creativity. Passionate about driving forward advancements in aerospace technology and fostering a spirit of innovation within the team.

DRDO, Ministry of Defence, Govt. of India

Summer R&D Intern

June 2024 - September 2024 (4 months)

Pune, Maharashtra, India

Currently engaged in the project titled "Manufacturing of Composite Structures for Aerospace Applications" project at RESEARCH & DEVELOPMENT ESTABLISHMENT (Engrs.), DRDO Pune's Composite Research Centre division. Conducting research on advanced composite materials and their aerospace applications. Assisting in the development and testing of composite structures to enhance performance and durability. Working under senior scientists and engineers to optimize manufacturing processes and contributing to innovative solutions for aerospace component design and fabrication.

International Centre for Theoretical Sciences

Research Fellow

June 2024 - July 2024 (2 months)

Bengaluru, Karnataka, India

Selected from a large applicant pool for the 2024 Summer School on Gravitational Wave Astronomy at the International Centre for Theoretical Science (ICTS-TIFR). Topics covered include the astrophysics of neutron stars and binaries (Dipankar Bhattacharya, Ashoka University), ultralight boson clouds around black holes (Nils Siemonsen, Princeton University), rapidly spinning neutron stars and emission mechanisms (Keith Riles, University of Michigan), and methods and results in searches for continuous gravitational waves (Alicia M Sintes, University of the Balearic Islands).

Google Developer Student Clubs

Head of Public Relations

August 2023 - July 2024 (1 year)

Mumbai, Maharashtra, India

In the role of Public Relations Head at GDSC - Amity University Mumbai, I elevate the voice of our tech community. Employing strategic communication,

I foster connections with stakeholders, curate impactful events, and nurture innovation. Empowering students to explore technology's forefront, I facilitate engaging experiences that promote growth and learning.

Atal Innovation Mission Official

2 years

Student Lead - STEM

June 2019 - May 2020 (1 year)

Solapur, Maharashtra, India

Led STEM initiatives at Atal Tinkering Lab, guiding students and organizing outreach programs to promote innovation and practical learning.

Student Innovator

June 2018 - June 2019 (1 year 1 month)

Solapur, Maharashtra, India

As an enthusiastic student innovator at the *Atal Tinkering Lab (ATL Lab)*, an initiative supported by the Government of India, I actively contributed to a wide range of STEM projects. These included developing an IoT-based Smart Dustbin System, Smart Home & City projects, Smart Agriculture Systems, Self-balancing robots, Rubik's Cube solvers, Line Follower Bots, and a Cubesat Model.

My hands-on experience extended to various skills, including coding and 3D printing. I was honored with the ATL Student Innovator 2020 Award by AIM, Government of India, Niti Aayog, and was nominated for the National Child Award 2022 in the Innovations category by the District Magistrate. This journey allowed me to apply innovation and creativity to real-world challenges, fostering a passion for STEM and technology.

Education

Amity University Mumbai

Bachelor of Technology - BTech, Aerospace, Aeronautical and Astronautical/
Space Engineering · (September 2022 - June 2026)

WALCHAND COLLEGE OF ARTS AND SCIENCE ASHOK CHOWK SOLAPUR DIST-SOLAPUR

Maharashtra Board (HSC) Class 12th Graduate, Science · (July 2020 - July 2022)

KLE English Medium School Solapur

CBSE class 10th Graduate · (June 2008 - June 2020)

Columbia University School of Professional Studies

Summer Immersion Pre-college Program, Entrepreneurship & Innovation |

Physics & Technologies of the future · (July 2021 - August 2021)